PROBLEM VF4

Pipe with a Partly Inserted Bar

Original statement · VF4.pdf

Slender flow criterion | Pipe and annulus in series | Matching the pressure

OFFICIAL PROBLEM STATEMENT

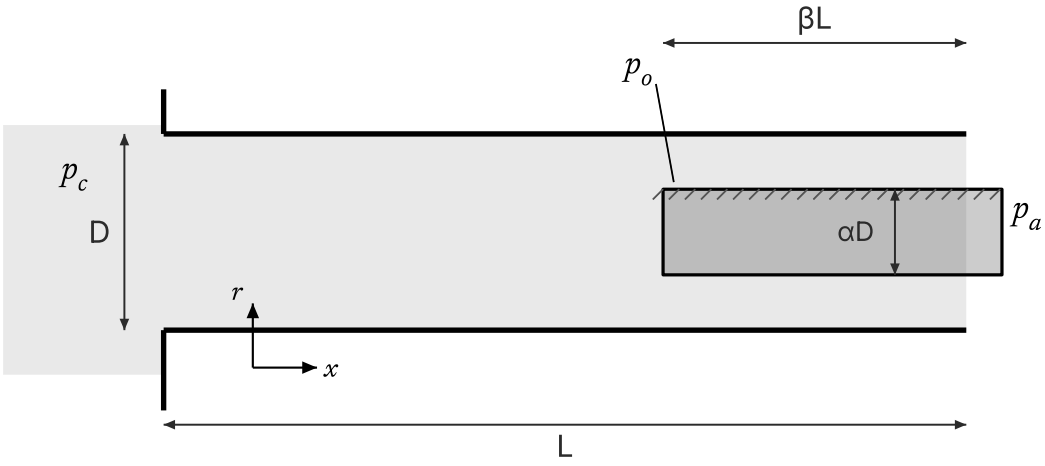
The liquid located in a container at pressure p_c discharges to the atmosphere, where the pressure is $p_a < p_c$, through a cylindrical pipe of diameter D and length $L \gg D$. To control the volume flux Q that circulates, a cylindrical bar of diameter αD is inserted coaxially from the exit, as shown in the figure, to partially block the liquid flow in the final stretch of the pipe.

1. Give the criterion for the motion in the tube to be dominated by viscosity.

2. Determine, in terms of the value of the pressure at the end of the tube p_o, the velocity field found for $0 < x/L < (1-\beta)$ and for $(1-\beta) < x/L < L$, where βL is the insertion length.

3. Obtain the volume flux Q and the value of p_o as a function of β.

On the sheet, "contained in a deposit" is struck out by hand and replaced by "located in a container", and the printed second range is written $(1-\beta) < x/L < L$, a typing slip for $(1-\beta) < x/L < 1$.



Statement figure (redrawn from VF4.pdf): a container at p_c discharging through a pipe of diameter D and length L; a coaxial bar of diameter αD enters from the exit along the last stretch βL; p_o is the pressure where the bar starts and p_a the exit pressure

Source: VF4.pdf (page 1, statement and handwritten key). Notes.pdf, Chapter 7: slender flows dominated by viscosity, Eqs. 7.40-7.46; pipe of variable section, Eqs. 7.47-7.51; entrance region, Eqs. 7.55-7.63.

Worked solution

Hypotheses, Data and Unknowns

- The liquid has constant ρ and μ; the flow is steady and axisymmetric.
- If the criterion of Part 1 holds, the flow is fully developed (Poiseuille) everywhere except in short regions of length of order D at the entrance and at the junction $x = (1-\beta)L$, and the pressure variations in them are small compared with the loss along the pipe (Notes.pdf, Eqs. 7.56 and 7.63). So $P(0) = p_c$ and the pressure at the end of the first stretch equals the one at the beginning of the second, p_o.
- Gravity is absorbed in the reduced pressure P; the gravity of a horizontal pipe does not matter, and the symbols p_c, p_o, p_a are used for the reduced pressures (as in the key).
- Let $a = D/2$, $\Delta p = p_c - p_a$ and $\mathcal{G} = \pi D^4/(128\mu)$, so that $\mathcal{G} = \pi a^4/(8\mu)$.

Symbol	Meaning	SI unit
D, αD	diameters of the pipe and of the bar	m
L, βL	length of the pipe and of the inserted bar	m
p_c, p_o, p_a	pressure in the container, at the bar tip, at the exit	Pa
Q	volume flux	m^3/s

Criterion for Viscous Dominance

The flow is dominated by viscosity when the convective acceleration is much smaller than the viscous force, Eq. (7.41), and the flow is steady here, so (7.40) is satisfied:

$$\frac{\rho v_L D}{\mu}\,\frac{D}{L} \ll 1$$

The velocity v_L is not known, so assume that viscosity dominates and estimate it from the balance of pressure and viscous forces, Eq. (7.44), $\Delta_L P \sim \mu v_L L/D^2$ with $\Delta_L P \sim p_c - p_a$:

$$v_L \sim \frac{D^2}{L}\,\frac{p_c - p_a}{\mu}$$

Substituting in the criterion:

$$\frac{\rho D}{\mu}\,\frac{D}{L}\,\frac{D^2}{L}\,\frac{p_c - p_a}{\mu} = \frac{\rho D^4\,(p_c - p_a)}{\mu^2 L^2} \ll 1$$

which only contains known data.

Velocity Fields

First stretch, $0 < x < (1-\beta)L$: circular pipe. As in (7.13)-(7.14), with a constant reduced pressure gradient $dP/dx = -P_{l1}$ and radius $a = D/2$:

$$v_x = \frac{P_{l1}}{4\mu}\left(a^2 - r^2\right) = \frac{P_{l1}}{16\mu}\left(D^2 - 4r^2\right)$$

The gradient is constant because the section is constant and Q is uniform, so $P_{l1} = (p_c - p_o)/\left[(1-\beta)L\right]$. In terms of the key's notation, $dP/dx = -(p_c - p_o)/\left[(1-\beta)L\right]$.

Second stretch, $(1-\beta)L < x < L$: annulus. The liquid is between the bar ($r = \alpha a$) and the pipe wall ($r = a$), both at rest, with $P_{l2} = (p_o - p_a)/(\beta L)$. Integrating (7.47) twice as in VF3, with $v_x(\alpha a) = v_x(a) = 0$:

$$v_x = \frac{P_{l2}}{4\mu}\left(a^2 - r^2\right) + C\ln\frac{r}{a}; \qquad 0 = \frac{P_{l2}}{4\mu}a^2\left(1-\alpha^2\right) + C\ln\alpha \implies C = -\frac{P_{l2}a^2\left(1-\alpha^2\right)}{4\mu\ln\alpha}$$

$$v_x = \frac{P_{l2}D^2}{16\mu}\left[1 - \left(\frac{2r}{D}\right)^2 - \frac{1-\alpha^2}{\ln\alpha}\ln\frac{2r}{D}\right], \qquad P_{l2} = \frac{p_o - p_a}{\beta L}$$

Check: at $r = D/2$ the bracket is $1 - 1 - 0 = 0$; at $r = \alpha D/2$ it is $1 - \alpha^2 - \frac{1-\alpha^2}{\ln\alpha}\ln\alpha = 0$.

Volume Flux

Pipe. From (7.15), $Q = \frac{\pi a^4}{8\mu} P_{l1}$, so

$$Q = \mathcal{G}\,\frac{p_c - p_o}{(1-\beta)L} \tag{1}$$

Annulus. With $s = r/a$, $Q = \int_{\alpha a}^{a} 2\pi r\, v_x\, dr = 2\pi a^2\,\frac{P_{l2}a^2}{4\mu}\, I$ with $I = \int_\alpha^1 s\left[1 - s^2 - k\ln s\right] ds$ and $k = (1-\alpha^2)/\ln\alpha$. Barrow's rule for each term:

$$\int_\alpha^1 s\,ds = \frac{1-\alpha^2}{2}, \qquad \int_\alpha^1 s^3 ds = \frac{1-\alpha^4}{4}, \qquad \int_\alpha^1 s\ln s\,ds = \left[\frac{s^2}{2}\ln s - \frac{s^2}{4}\right]_\alpha^1 = -\frac{1-\alpha^2}{4} - \frac{\alpha^2}{2}\ln\alpha$$

$$I = \frac{1-\alpha^2}{2} - \frac{1-\alpha^4}{4} + k\left[\frac{1-\alpha^2}{4} + \frac{\alpha^2}{2}\ln\alpha\right] = \frac{1-\alpha^2}{2} - \frac{1-\alpha^4}{4} + \frac{\left(1-\alpha^2\right)^2}{4\ln\alpha} + \frac{\alpha^2\left(1-\alpha^2\right)}{2}$$

and since $\frac{1-\alpha^2}{2} + \frac{\alpha^2(1-\alpha^2)}{2} = \frac{1-\alpha^4}{2}$, $I = \frac{1}{4}\left[\left(1-\alpha^4\right) - \frac{(1-\alpha^2)^2}{\ln(\alpha^{-1})}\right]$. Therefore

$$Q = \frac{\pi a^4 P_{l2}}{8\mu}\, f(\alpha) = \mathcal{G}\,\frac{p_o - p_a}{\beta L}\, f(\alpha), \qquad f(\alpha) = 1 - \alpha^4 - \frac{\left(1-\alpha^2\right)^2}{\ln\left(\alpha^{-1}\right)} \tag{2}$$

(f is the ratio of the flux of the annulus to that of the empty pipe for the same P_l: $f \to 1$ for $\alpha \to 0$ and $f \to 0$ for $\alpha \to 1$.) The key writes the same function as $f(\alpha) = 1 - \alpha^2(2 - \alpha^2) - \frac{(1-\alpha^2)\left[1-\alpha^2-2\alpha^2\ln(\alpha^{-1})\right]}{\ln(\alpha^{-1})}$; expanding, $1 - 2\alpha^2 + \alpha^4 + 2\alpha^2(1-\alpha^2) - (1-\alpha^2)^2/\ln(\alpha^{-1}) = 1 - \alpha^4 - (1-\alpha^2)^2/\ln(\alpha^{-1})$, the same.

Flux and Pressure at the Junction

The flux is the same in both stretches (Notes.pdf, Eq. 7.72 and the discussion of (7.49)). Equating (1) and (2):

$$\frac{p_c - p_o}{1-\beta} = f\,\frac{p_o - p_a}{\beta}$$

With $u = p_o - p_a$ and $p_c - p_o = \Delta p - u$: $\beta(\Delta p - u) = f(1-\beta)u$, so $u = \dfrac{\beta\,\Delta p}{\beta + f(1-\beta)}$. Then

$$Q = \mathcal{G}\,\frac{f\,u}{\beta L} = \frac{\pi D^4}{128\mu L}\,\frac{p_c - p_a}{1 + \beta\,(1-f)\,/f}, \qquad p_o = p_c - \frac{(1-\beta)\,(p_c - p_a)}{1 + \beta\,(1-f)\,/f}$$

(the second from $p_c - p_o = \Delta p - u = \dfrac{f(1-\beta)\Delta p}{\beta + f(1-\beta)}$, dividing numerator and denominator by f).

Discrepancy with the official solution

The two boxed results of the key have $1 + \dfrac{\beta f}{1-f}$ in the denominator, with $f = f(\alpha)$ of its own equation (2). That combination is not the one that follows from (1) and (2). Two limits prove it. For $\alpha \to 0$ (no bar) $f \to 1$, and the key's denominator $1 + \beta f/(1-f)$ tends to infinity, so $Q \to 0$, when the flow must be that of the empty pipe; the corrected denominator $1 + \beta(1-f)/f$ tends to 1 and gives $Q = \pi D^4 \Delta p/(128\mu L)$. For $\alpha \to 1$ (the bar fills the pipe) $f \to 0$ and the key gives the empty-pipe flux, the corrected formula $Q \to 0$. The calculation above is kept.

Answers and Checks

- Viscous dominance: $\rho D^4 (p_c - p_a)/(\mu^2 L^2) \ll 1$.
- Velocities: $v_x = \frac{(p_c - p_o)}{16\mu(1-\beta)L}\left(D^2 - 4r^2\right)$ for $0 < x < (1-\beta)L$ and $v_x = \frac{(p_o - p_a)D^2}{16\mu\beta L}\left[1 - (2r/D)^2 - \frac{1-\alpha^2}{\ln\alpha}\ln(2r/D)\right]$ for $(1-\beta)L < x < L$.
- Flux: $Q = \frac{\pi D^4}{128\mu L}\frac{p_c - p_a}{1+\beta(1-f)/f}$; pressure at the junction: $p_o = p_c - \frac{(1-\beta)(p_c - p_a)}{1+\beta(1-f)/f}$.

- Limits: $\beta \to 0$ (no bar) gives $Q = \pi D^4 \Delta p / (128 \mu L)$ and $p_o \to p_c - \Delta p = p_a$; $\beta \to 1$ (bar over the whole pipe) gives $Q = \mathcal{G} f \, \Delta p / L$ and $p_o \to p_c$, all the loss in the annulus; $f \to 1$ gives the empty pipe whatever β.
- Dimensions: $\pi D^4 \Delta p / (\mu L)$ is $L^4 \cdot M L^{-1} T^{-2} / (M L^{-1} T^{-1} \cdot L) = L^3 T^{-1}$, a volume flux; $\rho D^4 \Delta p / (\mu^2 L^2)$ is $M L^{-3} L^4 \, M L^{-1} T^{-2} / (M^2 L^{-2} T^{-2} L^2) = 1$.
- Check against the official key: VF4.pdf gives the same criterion, the same fields, the same flux in each stretch (1) and (2) and the same $f(\alpha)$; its last two boxed results differ from the corrected ones, as explained in the discrepancy above.